Capstone 2 Final Report: Airline Delay Prediction

1. Problem Statement

The goal of this project is to predict arrival delays using airline delay data and identify key factors that contribute to delays. Understanding these factors can help improve operational efficiency and reduce overall delays.

2. Data & Preprocessing

The dataset was cleaned and prepared for modeling. The target variable (arr_delay) was defined, and categorical variables were encoded using one-hot encoding. Missing values were handled, and all features were converted to numeric format to ensure compatibility with machine learning models.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand relationships between features and arrival delay. The analysis showed that delay-related factors such as late aircraft, carrier, and NAS delays were the largest contributors to total delay. These findings guided the modeling approach.

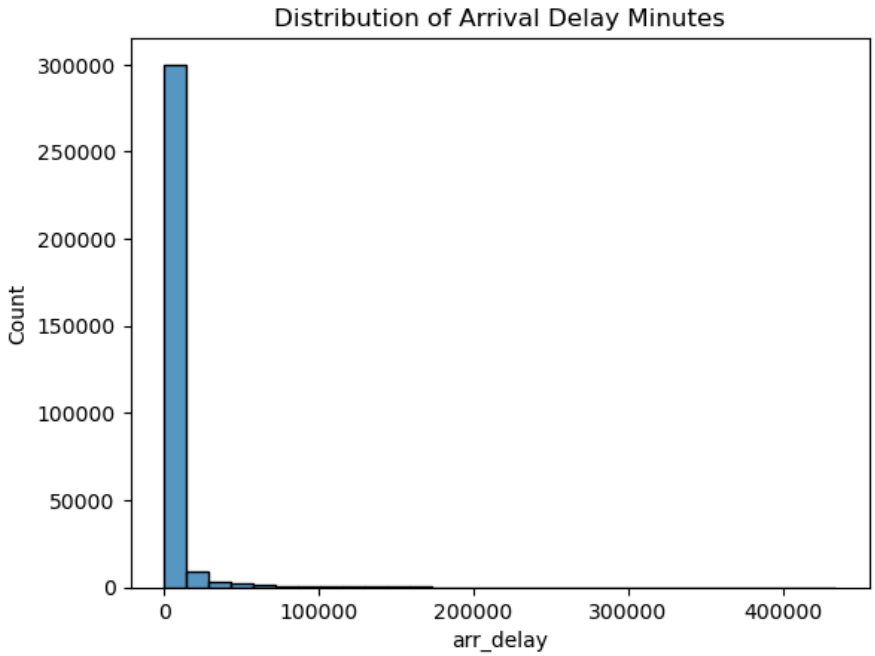

Figure 1: Distribution of Arrival Delay Minutes

The distribution of arrival delay minutes is highly right-skewed, with most flights experiencing relatively small delays and a small number of extreme values. This indicates the presence of outliers and suggests that models must account for variability and non-linear patterns in delay behavior.

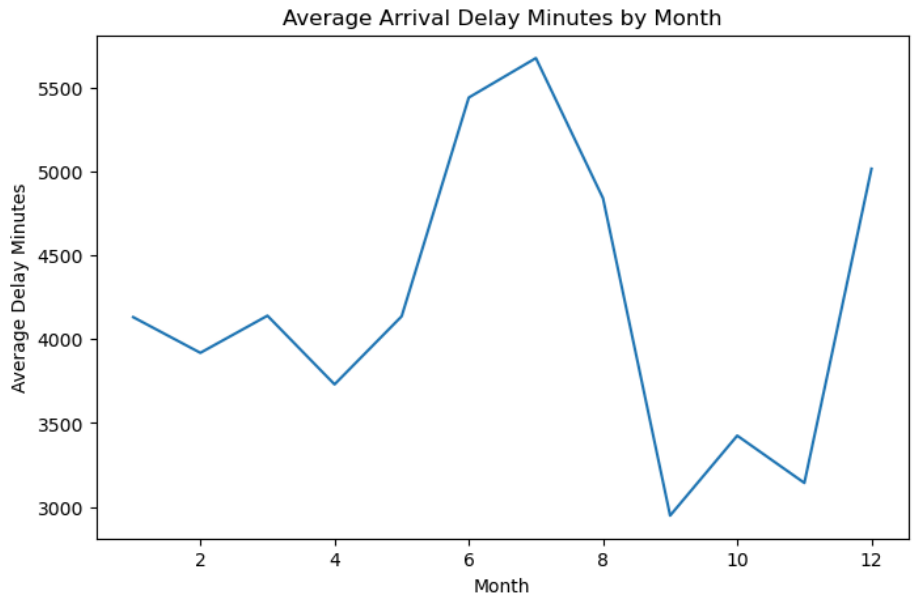

Figure 2: Average Arrival Delay by Month

Average arrival delay minutes peak during the summer months (June–August) and decline significantly in early fall. This pattern suggests that seasonal factors such as increased travel demand and weather conditions contribute to higher delays.

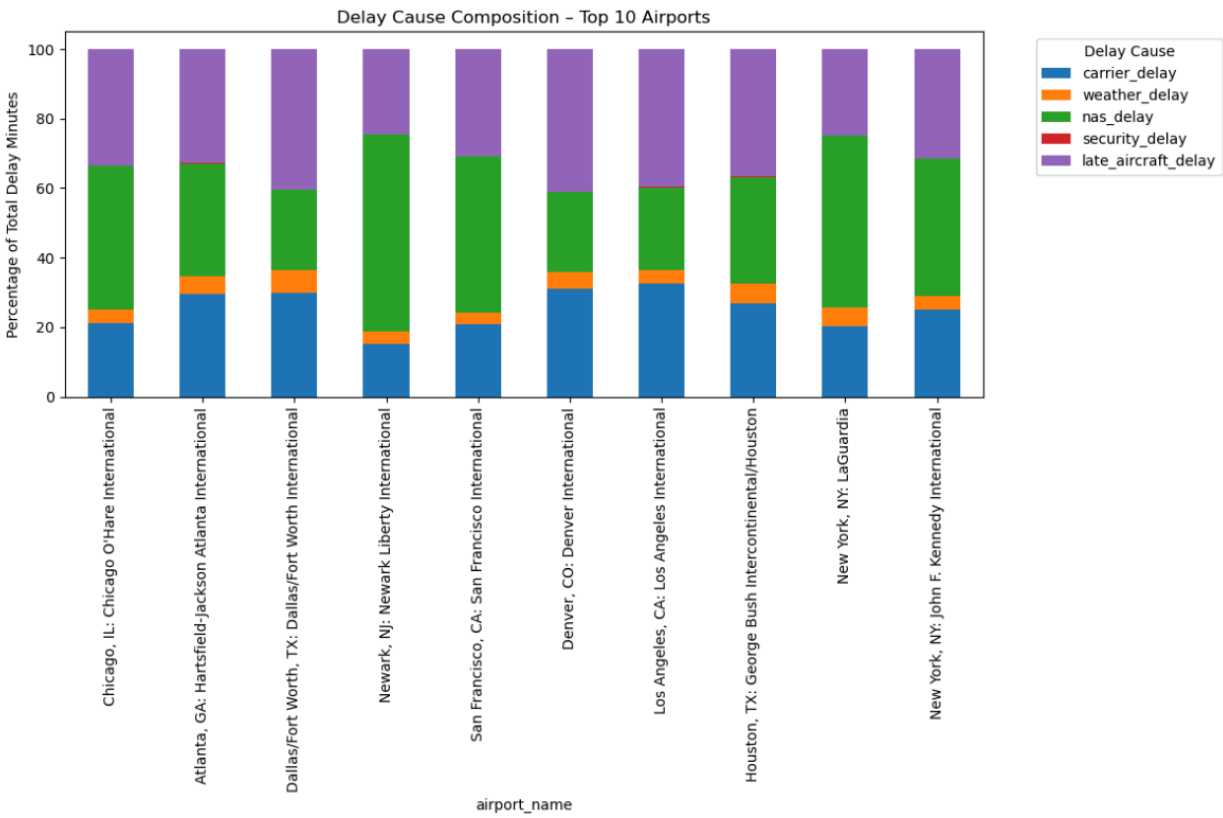

Figure 3: Delay Cause Composition – Top Airports

Late aircraft and NAS delays are the dominant contributors across major airports, with carrier delays also playing a significant role. This highlights how delays propagate through the system and supports focusing on upstream operational improvements.

4. Modeling Approach

The dataset was split into training and testing sets to evaluate performance on unseen data. Feature scaling was applied for linear models, with the scaler fit on training data only to prevent data leakage. Three models were trained: Linear Regression as a baseline, Ridge Regression to address high dimensionality, and Random Forest to capture non-linear relationships.

5. Model Results

Linear Regression achieved an R^2 of 0.827 and RMSE of approximately 5192. Ridge Regression produced nearly identical results, indicating that regularization had minimal impact. Random Forest performed best, achieving an R^2 of 0.888 and RMSE of approximately 4179.

6. Final Model Selection

Random Forest was selected as the final model due to its higher R^2 and lower RMSE. This suggests that non-linear relationships play an important role in predicting arrival delays.

7. Recommendations

- Focus on reducing late aircraft delays, as they contribute the most to total delay.
- Improve scheduling efficiency to minimize cascading delays.
- Use predictive models to anticipate high-delay scenarios and allocate resources more effectively.

8. Conclusion / Future Work

This project demonstrates that arrival delays are influenced by multiple interacting factors. While linear models provided strong baseline performance, the Random Forest model was better suited for capturing more complex patterns. Future improvements could include feature engineering and hyperparameter tuning.